Shared vs. Independent Stratification Vector (f) Under Real LD Structure

A cross-scale, cross-chromosome comparison using 1000 Genomes chr1 and chr22 LD structure

Executive Summary

This analysis extends the project's AR1-based heritability simulation framework to use **real linkage-disequilibrium (LD) structure** drawn from 1000 Genomes chromosome 1 and chromosome 22 genotypes (in place of synthetic AR1 correlation), and tests how a reference panel's stratification vector f behaves when it is **shared** with the main sample versus drawn **independently**. The comparison was run across four sample/SNP scales (n=500/m=1000, n=1000/m=2000, n=2000/m=4000, n=5000/m=10000) and both chromosomes, for a total of 352 simulated datasets (11 F_{ST} values × 100 replicates × 4 scenarios × 4 scales × 2 chromosomes).

Headline finding: without a reference panel, whether f is shared or redrawn per replicate makes almost no difference to the LDSC free-intercept h^2 estimate (gap < 0.05 at F_{ST}=0.5, flipping sign across scales - noise, not a real effect). **With a reference panel**, however, an f-mismatched reference panel produces estimates that increasingly diverge from the shared-f case as F_{ST} grows - and this divergence gets **substantially worse** as sample size and SNP count scale up, not better. The gap at F_{ST}=0.5 grows roughly monotonically from ~0.36-0.49 at n=500 to ~3.95-4.08 at n=5000, and the pattern is essentially identical between chr1 and chr22, indicating this is a property of sample size rather than of which chromosome's LD structure is used.

This is a different failure mode than was seen in the project's earlier AR1-based reference-panel work: there, an f-mismatched reference panel caused the free-intercept estimator to *collapse toward zero*. Under real LD structure, the same mismatch instead causes the estimator to *inflate*, and the inflation grows with scale.

Methods

Genotypes were simulated with LD structure taken from real 1000 Genomes EUR chr1 (106,136 SNPs) and chr22 (18,613 SNPs) PLINK files, using the data_generation class's realistic=True path: synthetic individuals are drawn as random linear combinations of the real genotype matrix's columns, preserving its LD structure while allowing arbitrary sample size. A genetic-drift/stratification vector f (parameterized by F_{ST}) splits the sample into two subpopulations, exactly as in the project's AR1 pipeline.

Four scenarios were run at every scale and chromosome:

- **No ref panel, shared f** - LD scores computed in-sample; f drawn once per F_{ST} and reused across all 100 replicates.
- **No ref panel, independent f** - LD scores computed in-sample; f redrawn fresh every replicate.
- **Ref panel, shared f** - LD scores/GWASH statistics taken from a separately-simulated reference panel; the reference panel's f is shared with the main replicates.
- **Ref panel, independent f** - same reference panel LD scores, but the main replicates draw their own f, independent of (mismatched with) the reference panel's.

The reported estimator is LDSC with a free intercept (h2_ldsc_reg), averaged over 100 replicates per F_{ST} value ($F_{ST} \in$ {0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5}). A theoretical reference line h2_pop + $sigma_s^2/C_f$ (C_f = mean($f^2/(1+f^2)$)) is overlaid on each plot; this formula was numerically validated against the project's R implementation (Sam_R_code_fast_randf.R) under the original AR1 setting, agreeing to within 0.7% across the F_{ST} grid - well within sampling noise from 100 replicates.

Fixed parameters throughout: h2_pop=0.3, $sigma_s$=0.3, seed=123. m is always 2n.

Summary: h2 estimate gap (independent f − shared f) at F_ST=0.5

Mean LDSC free-intercept h2 estimate over 100 replicates at F_{ST}=0.3 and F_{ST}=0.5, for shared-f and independent-f, in both the no-ref-panel and ref-panel settings.

Chrom	n	m	Setting	Shared f (F_ST=0.3)	Indep f (F_ST=0.3)	Shared f (F_ST=0.5)	Indep f (F_ST=0.5)	Gap (F_ST=0.5)
chr1	500	1000	ref panel	0.541	0.605	0.508	0.869	+0.362
chr1	500	1000	no ref panel	0.514	0.490	0.447	0.422	-0.025
chr1	1000	2000	ref panel	0.526	1.052	0.412	1.512	+1.100
chr1	1000	2000	no ref panel	0.478	0.489	0.428	0.422	-0.006
chr1	2000	4000	ref panel	0.346	1.726	0.317	2.532	+2.216
chr1	2000	4000	no ref panel	0.451	0.427	0.396	0.377	-0.020
chr1	5000	10000	ref panel	0.515	3.037	0.449	4.529	+4.080
chr1	5000	10000	no ref panel	0.396	0.433	0.343	0.386	+0.043
chr22	500	1000	ref panel	0.525	0.656	0.478	0.967	+0.490
chr22	500	1000	no ref panel	0.568	0.529	0.490	0.470	-0.020
chr22	1000	2000	ref panel	0.491	1.026	0.429	1.478	+1.049
chr22	1000	2000	no ref panel	0.485	0.450	0.430	0.388	-0.042
chr22	2000	4000	ref panel	0.349	1.716	0.334	2.481	+2.147
chr22	2000	4000	no ref panel	0.454	0.432	0.399	0.382	-0.017
chr22	5000	10000	ref panel	0.510	2.942	0.442	4.394	+3.952
chr22	5000	10000	no ref panel	0.392	0.426	0.341	0.382	+0.041

Note the no-ref-panel gap stays small (<0.05 in magnitude, sign-flipping) regardless of scale - consistent with noise. The ref-panel gap grows from ~0.36-0.49 at n=500 to ~3.95-4.08 at n=5000, roughly an order of magnitude, and tracks almost identically between chr1 and chr22 at every scale.

Summary: h2 estimate at low F_ST (F_ST = 0, 0.005, 0.01)

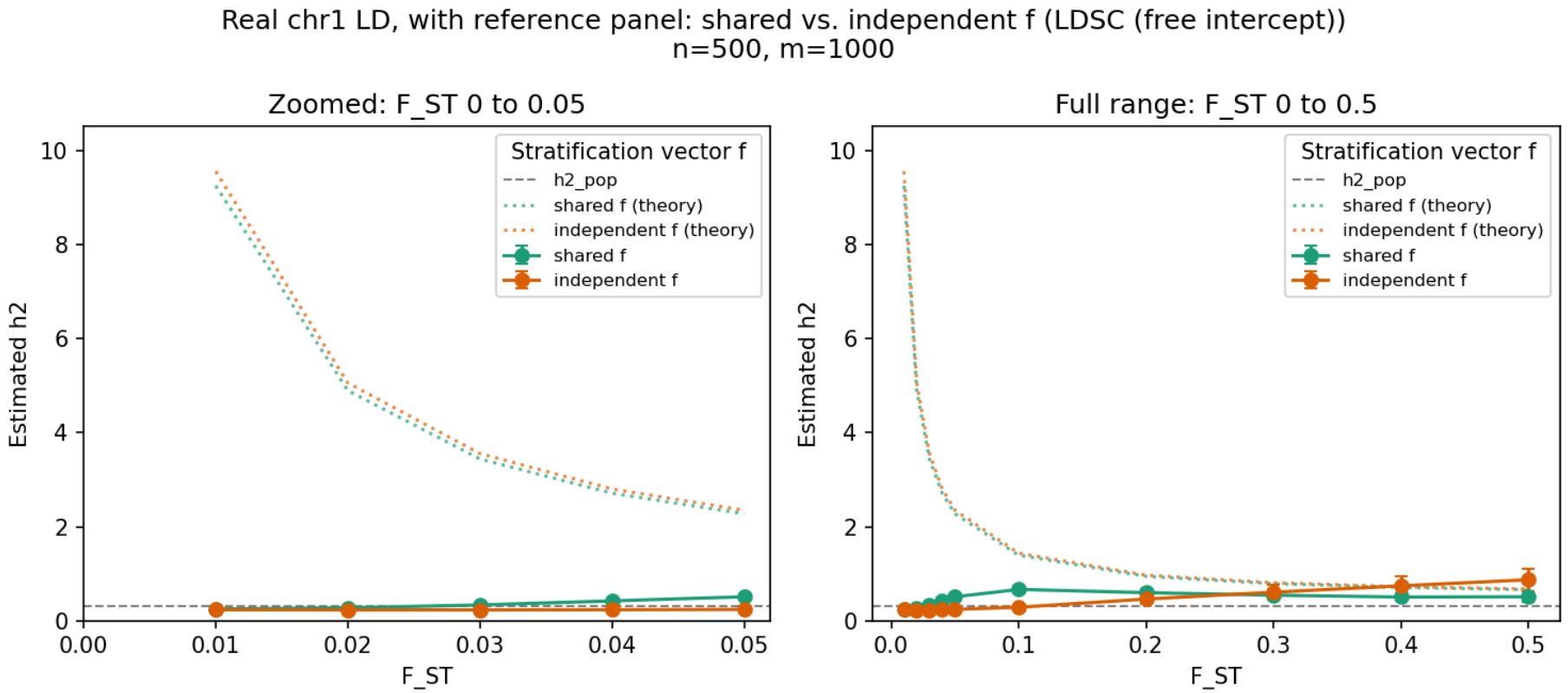
Mean LDSC free-intercept h2 estimate over 100 replicates at F_{ST}=0, F_{ST}=0.005, and F_{ST}=0.01, for shared-f and independent-f, in both the no-ref-panel and ref-panel settings. Note F_{ST}=0.005 is not part of this pipeline's regular grid (0, 0.01, 0.02, ..., 0.5) and was generated specifically for this comparison.

Chrom	n	Setting	Shared f (F_ST=0)	Indep f (F_ST=0)	Shared f (F_ST=0.005)	Indep f (F_ST=0.005)	Shared f (F_ST=0.01)	Indep f (F_ST=0.01)
chr1	500	ref panel	0.256	0.239	0.247	0.233	0.246	0.231
chr1	500	no ref panel	0.273	0.260	0.261	0.255	0.259	0.259
chr1	1000	ref panel	0.253	0.253	0.237	0.249	0.243	0.246
chr1	1000	no ref panel	0.260	0.263	0.250	0.258	0.256	0.270
chr1	2000	ref panel	0.224	0.242	0.228	0.235	0.261	0.239
chr1	2000	no ref panel	0.228	0.247	0.224	0.241	0.249	0.263
chr1	5000	ref panel	0.192	0.192	0.232	0.223	0.322	0.277
chr1	5000	no ref panel	0.195	0.195	0.228	0.229	0.304	0.308
chr22	500	ref panel	0.303	0.288	0.307	0.287	0.324	0.284
chr22	500	no ref panel	0.302	0.293	0.300	0.292	0.310	0.300
chr22	1000	ref panel	0.262	0.256	0.256	0.250	0.264	0.247
chr22	1000	no ref panel	0.275	0.270	0.274	0.264	0.285	0.274
chr22	2000	ref panel	0.224	0.229	0.224	0.226	0.251	0.233
chr22	2000	no ref panel	0.231	0.237	0.226	0.235	0.246	0.259
chr22	5000	ref panel	0.195	0.189	0.234	0.224	0.313	0.276
chr22	5000	no ref panel	0.196	0.191	0.224	0.226	0.288	0.293

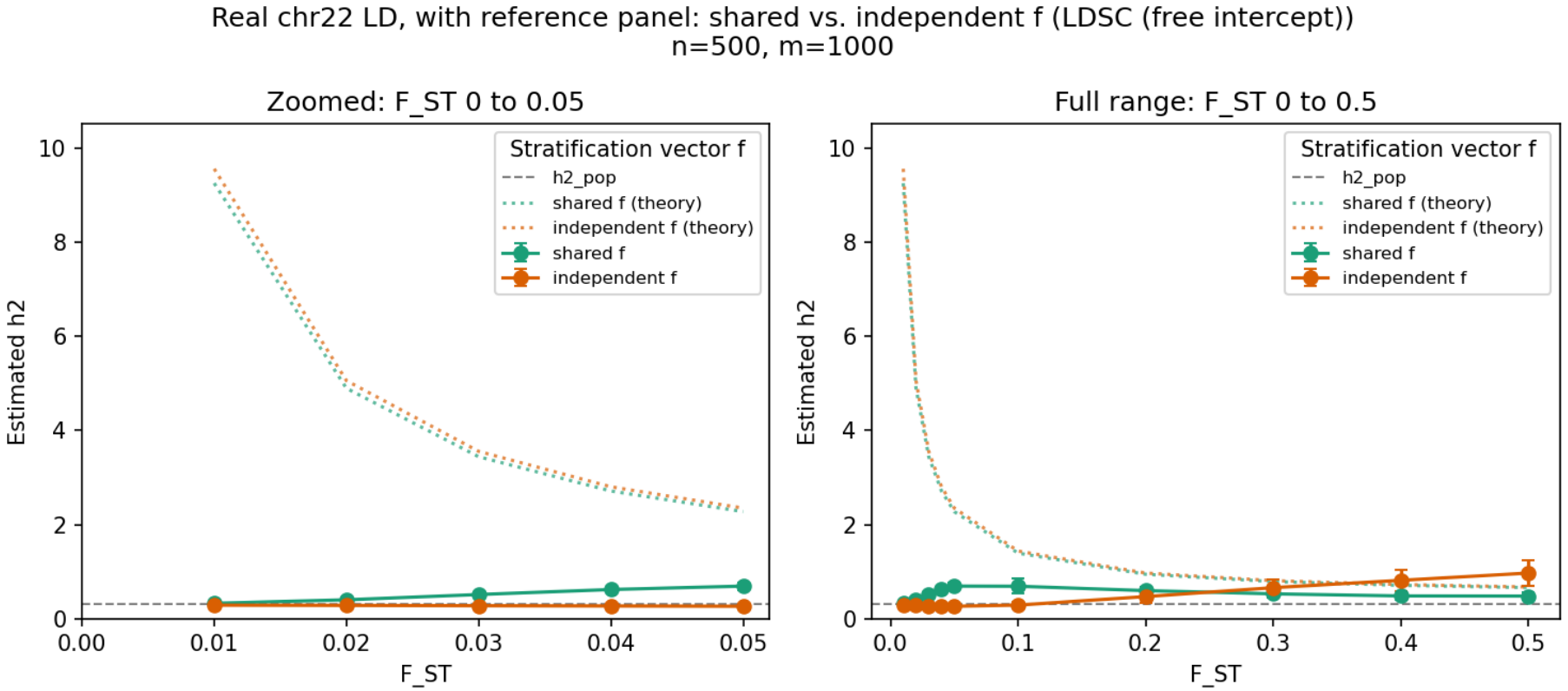
Shared-f and independent-f are statistically indistinguishable at every scale, chromosome, and panel setting across this low-F_ST range - differences are within a few thousandths and do not consistently favor one direction. This confirms the divergence seen at F_ST=0.3/0.5 is a genuine F_ST-driven effect, not an artifact present from F_ST=0.

Figures: Reference-Panel Setting (the main finding)

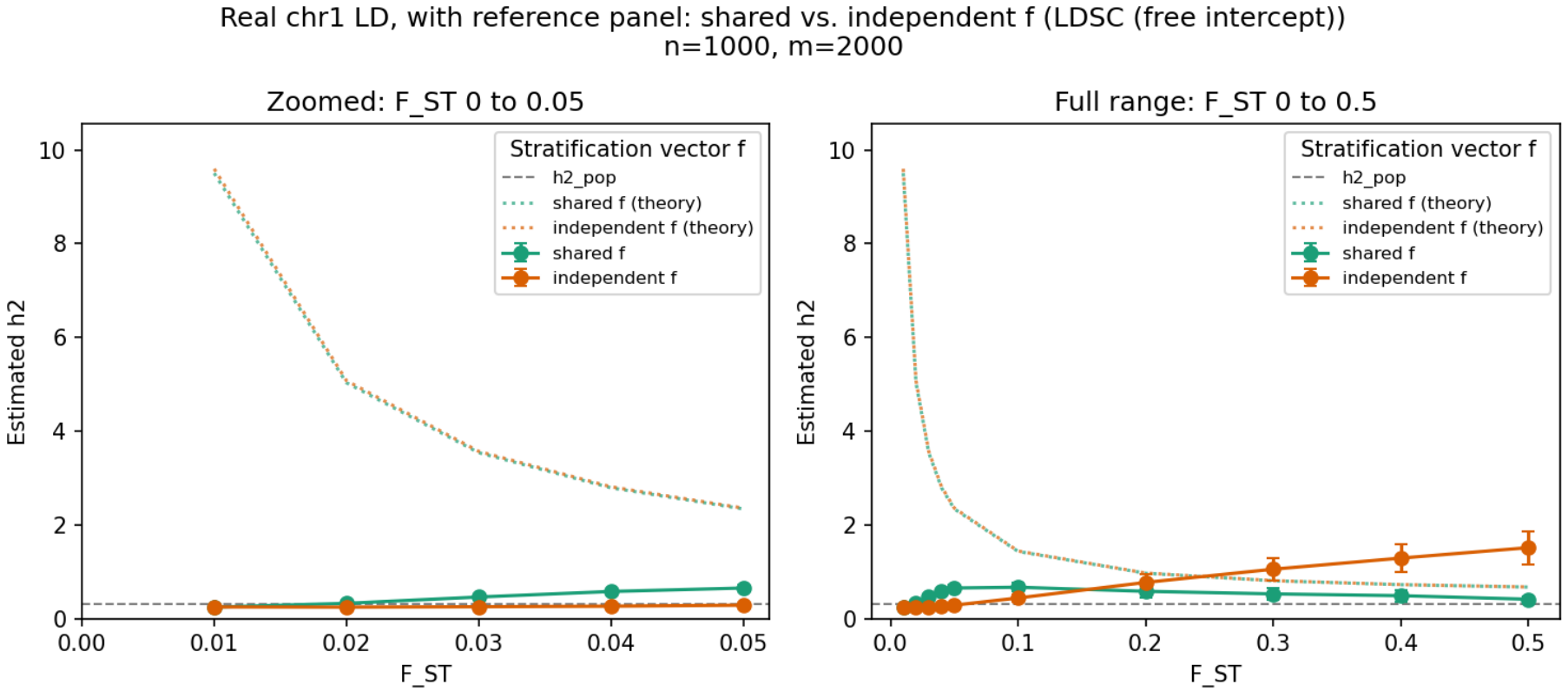
Each figure shows mean LDSC free-intercept h2 (±95% CI) vs. F_{ST}, for shared f (green) vs. independent f (orange), with the theoretical line overlaid (dotted). Left panel zooms into F_ST 0-0.05; right panel shows the full 0-0.5 range.



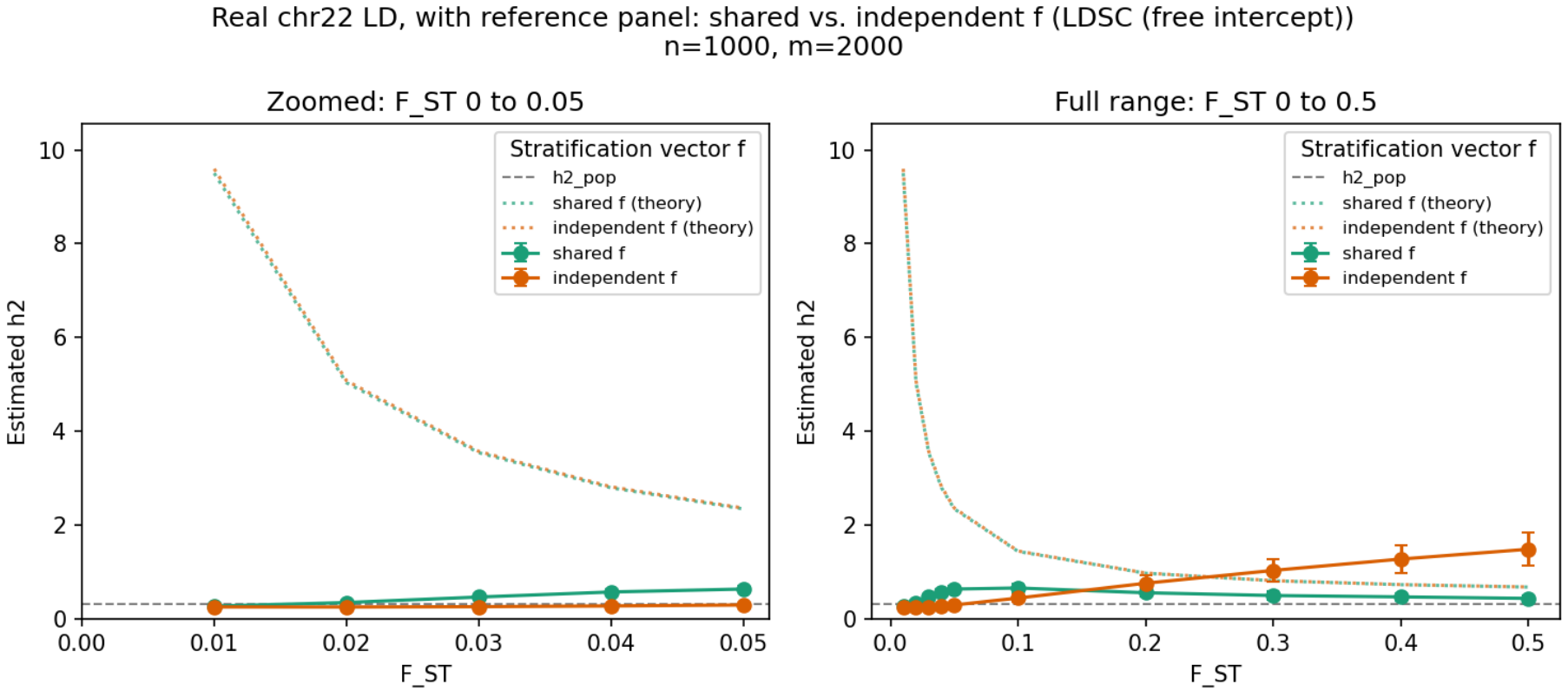
chr1, n=500, m=1000 — reference panel setting



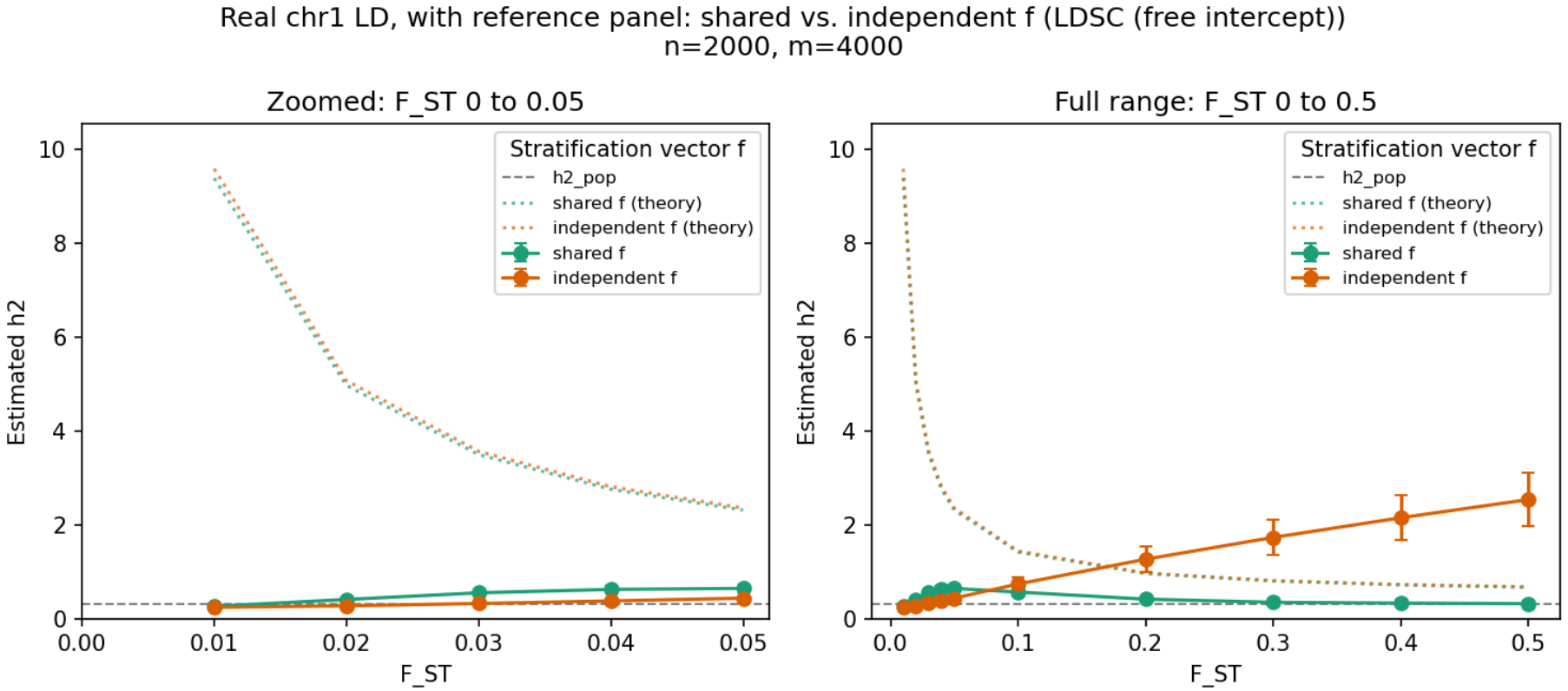
chr22, n=500, m=1000 — reference panel setting



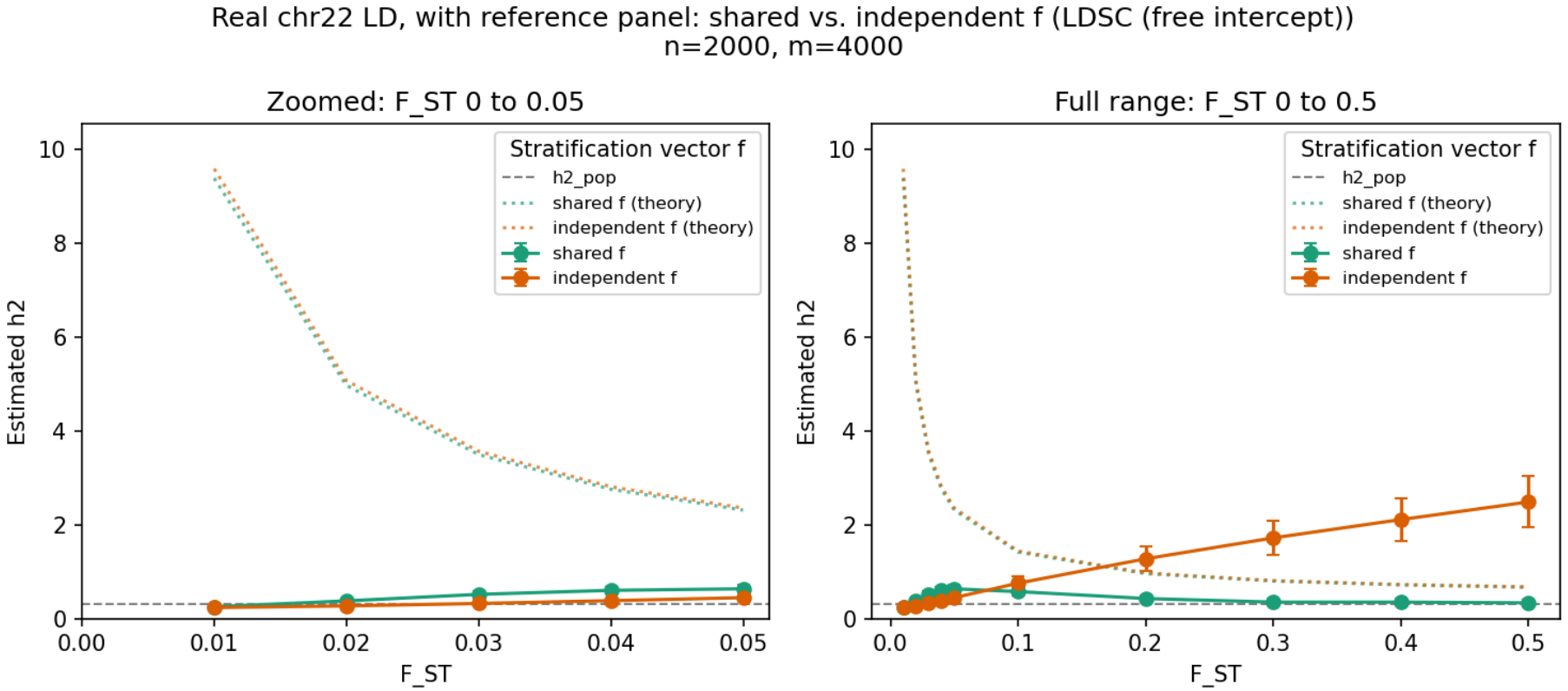
chr1, n=1000, m=2000 — reference panel setting



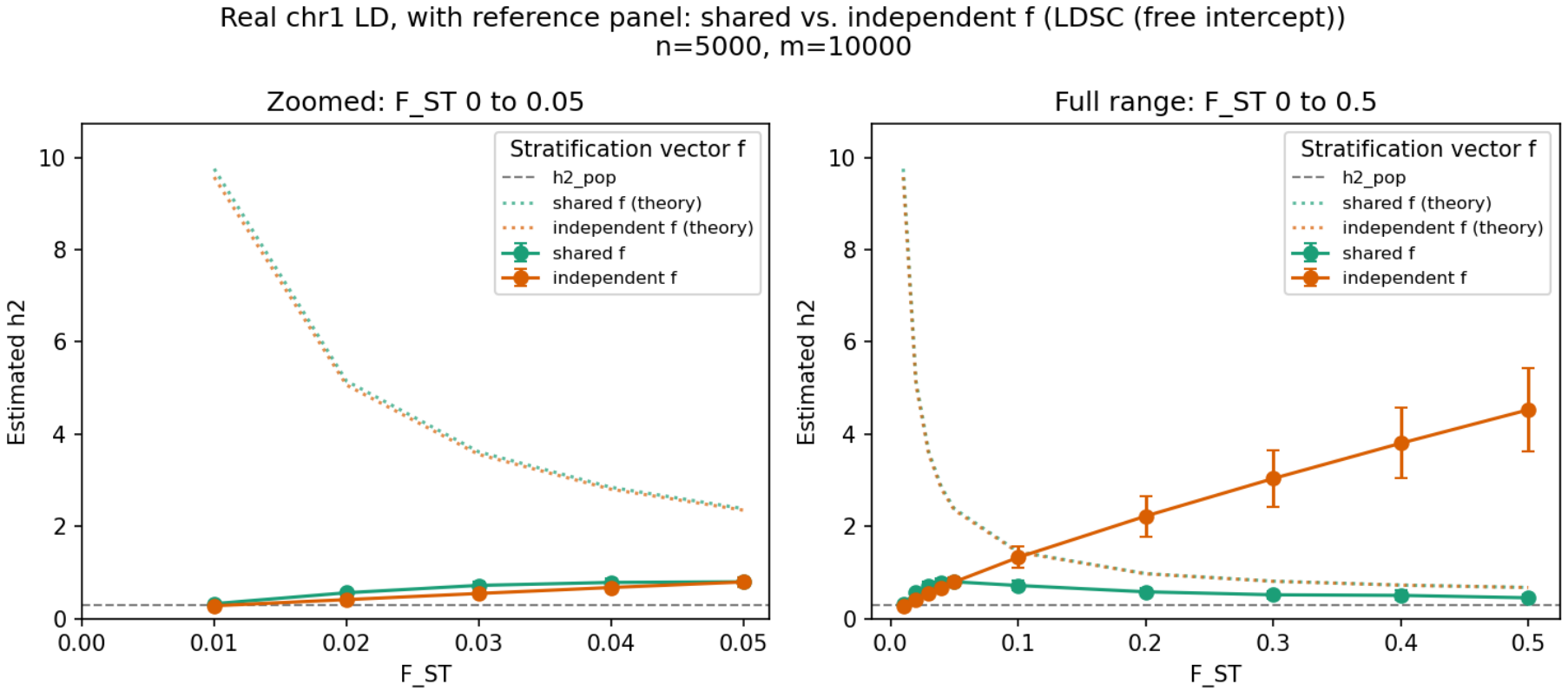
chr22, n=1000, m=2000 — reference panel setting



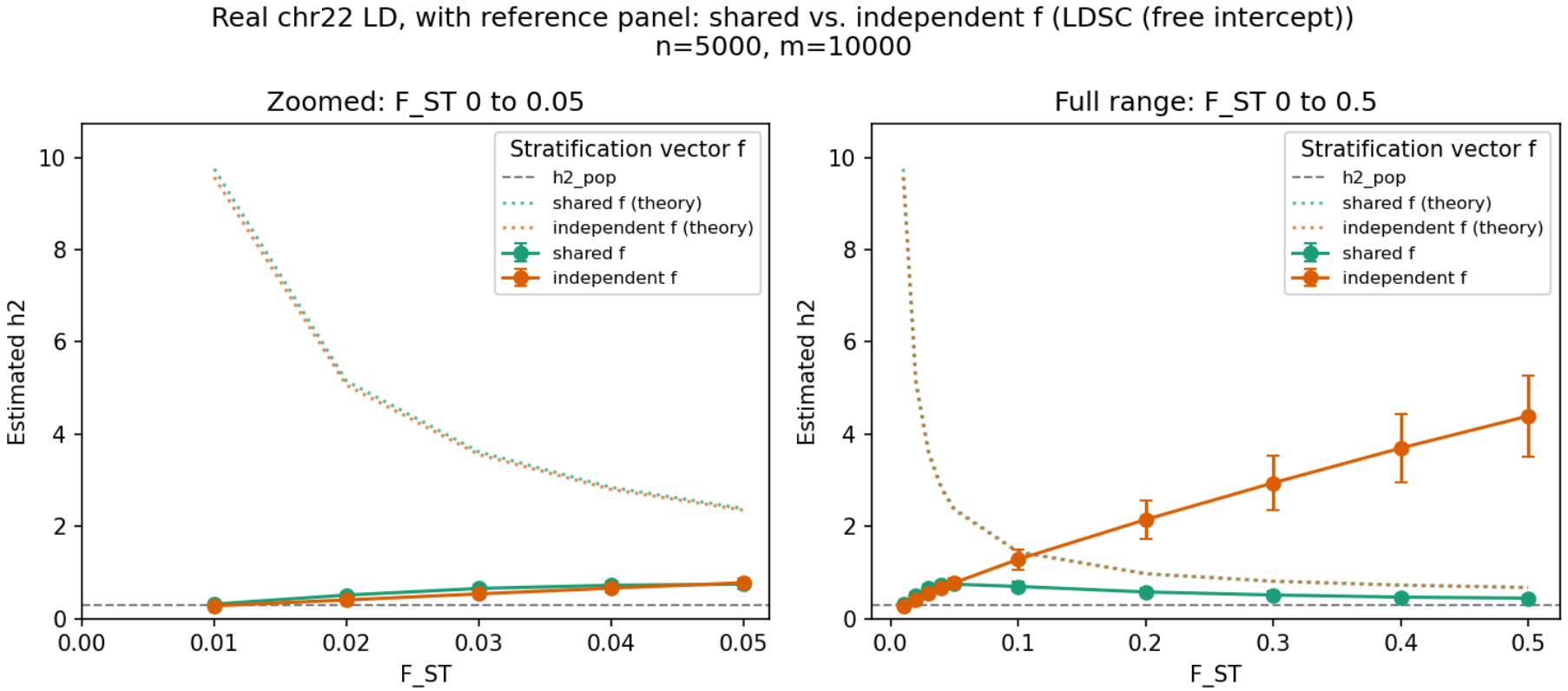
chr1, n=2000, m=4000 — reference panel setting



chr22, n=2000, m=4000 — reference panel setting



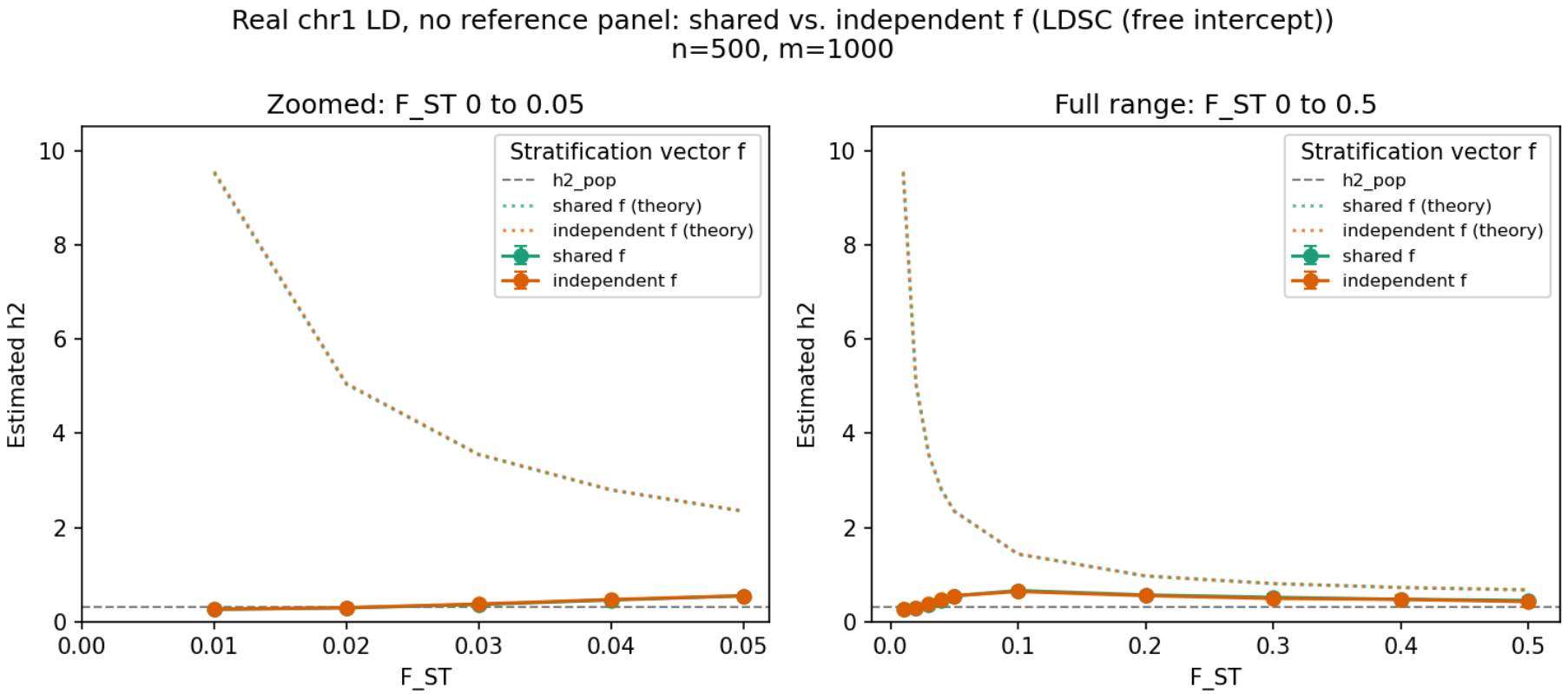
chr1, n=5000, m=10000 — reference panel setting



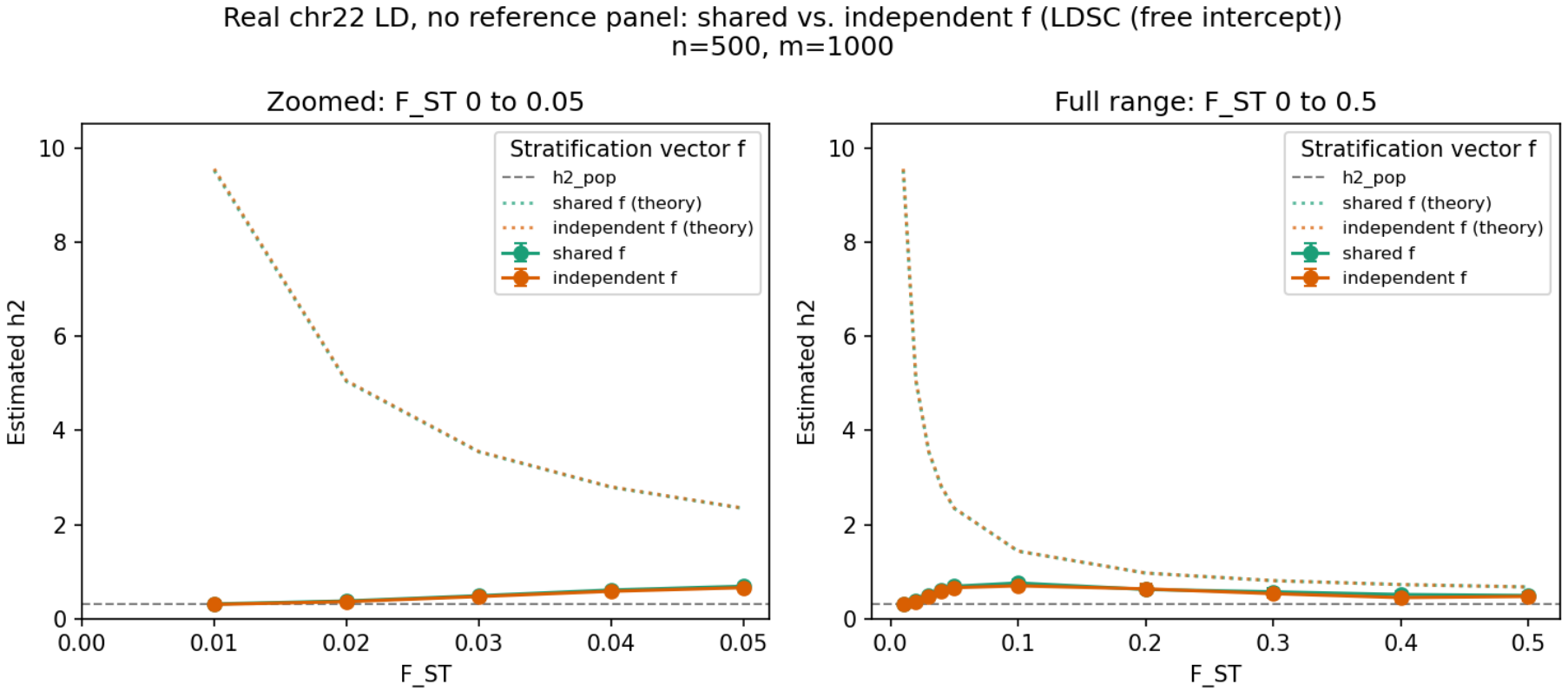
chr22, n=5000, m=10000 — reference panel setting

Figures: No-Reference-Panel Setting (for contrast)

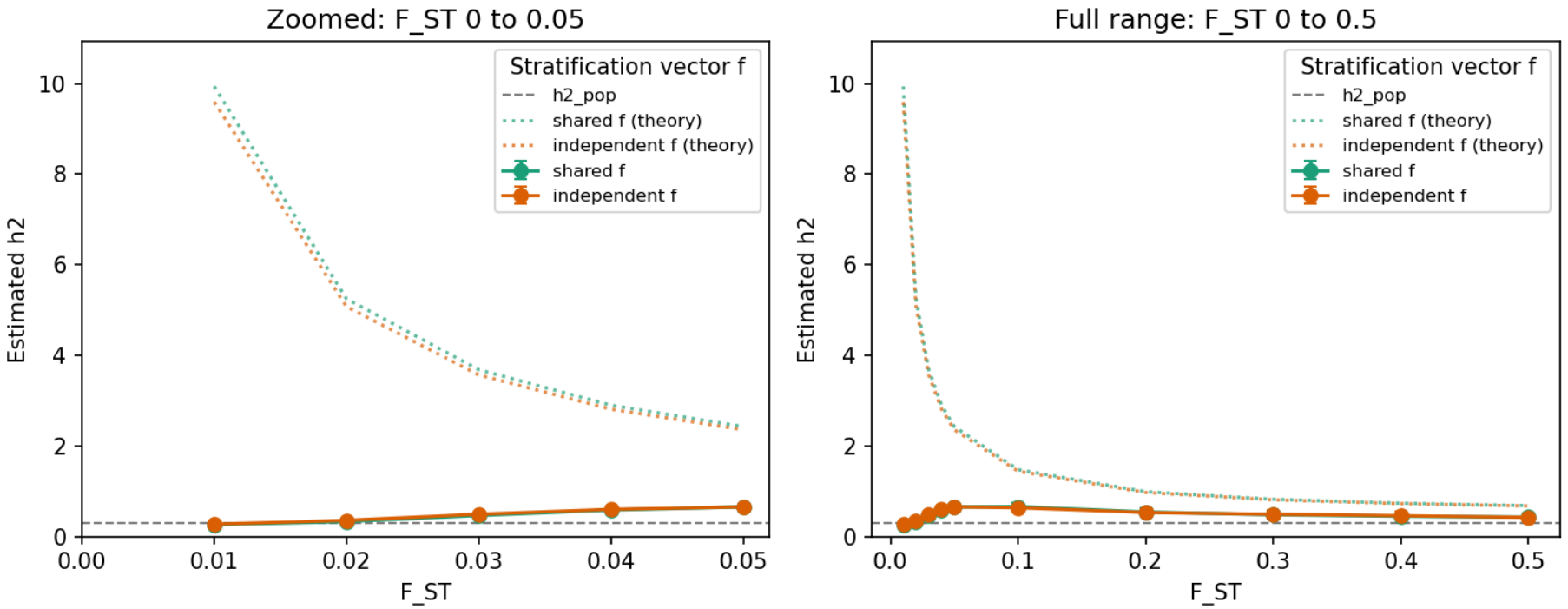
Shown at every scale on both chromosomes, confirming shared-f and independent-f track each other closely throughout when LD scores are computed in-sample (no reference panel involved).



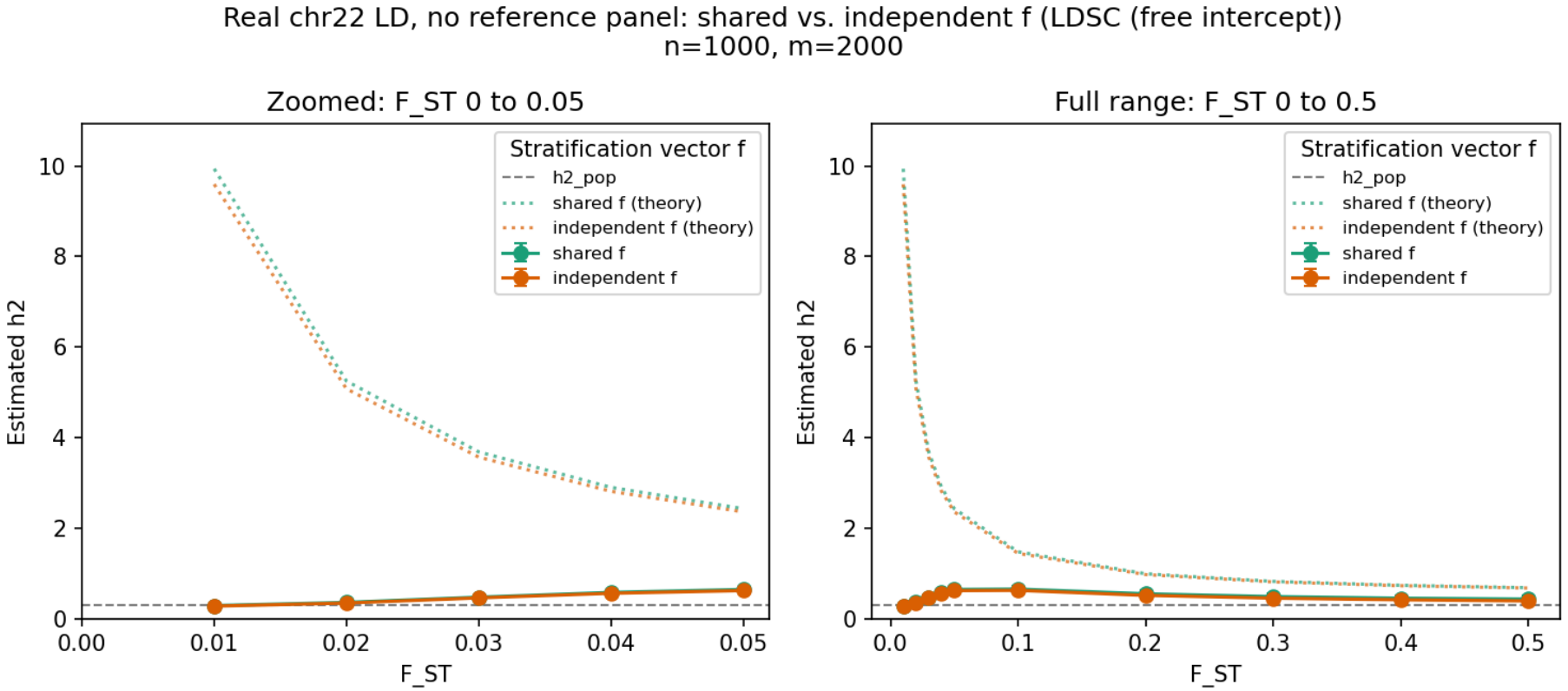
chr1, n=500, m=1000 — no reference panel



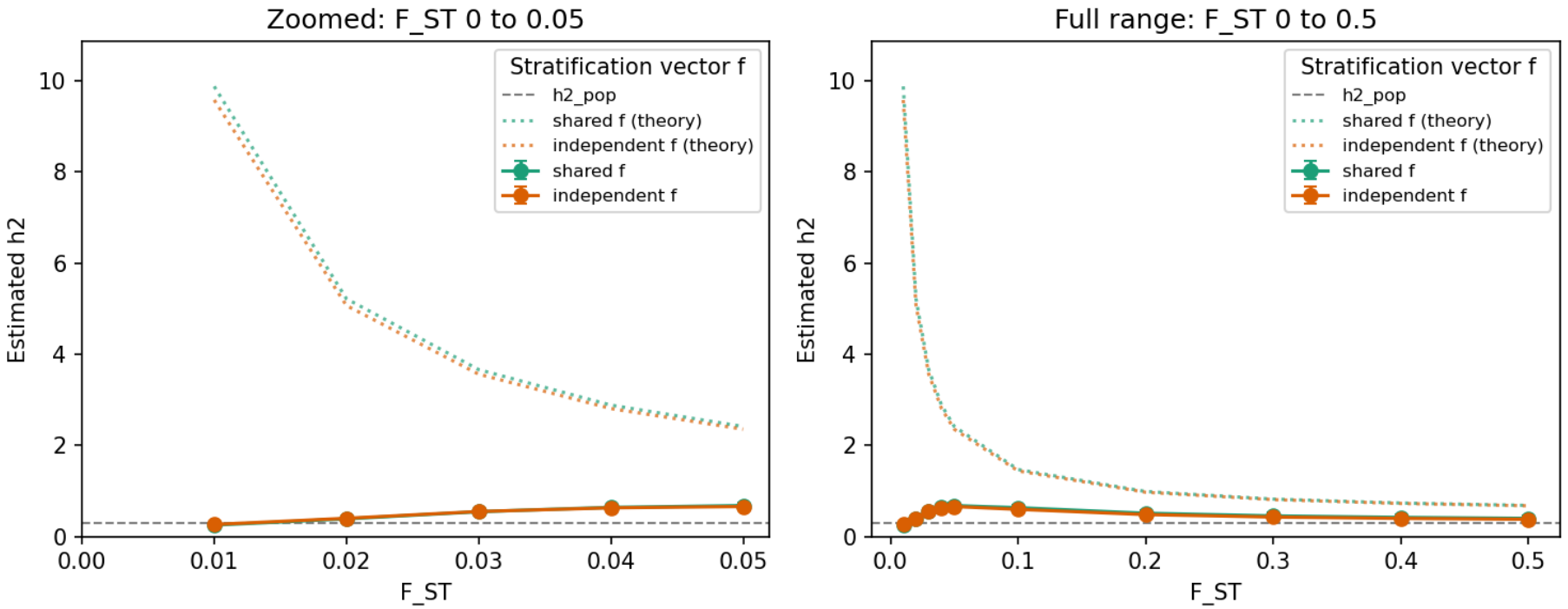
chr22, n=500, m=1000 — no reference panel



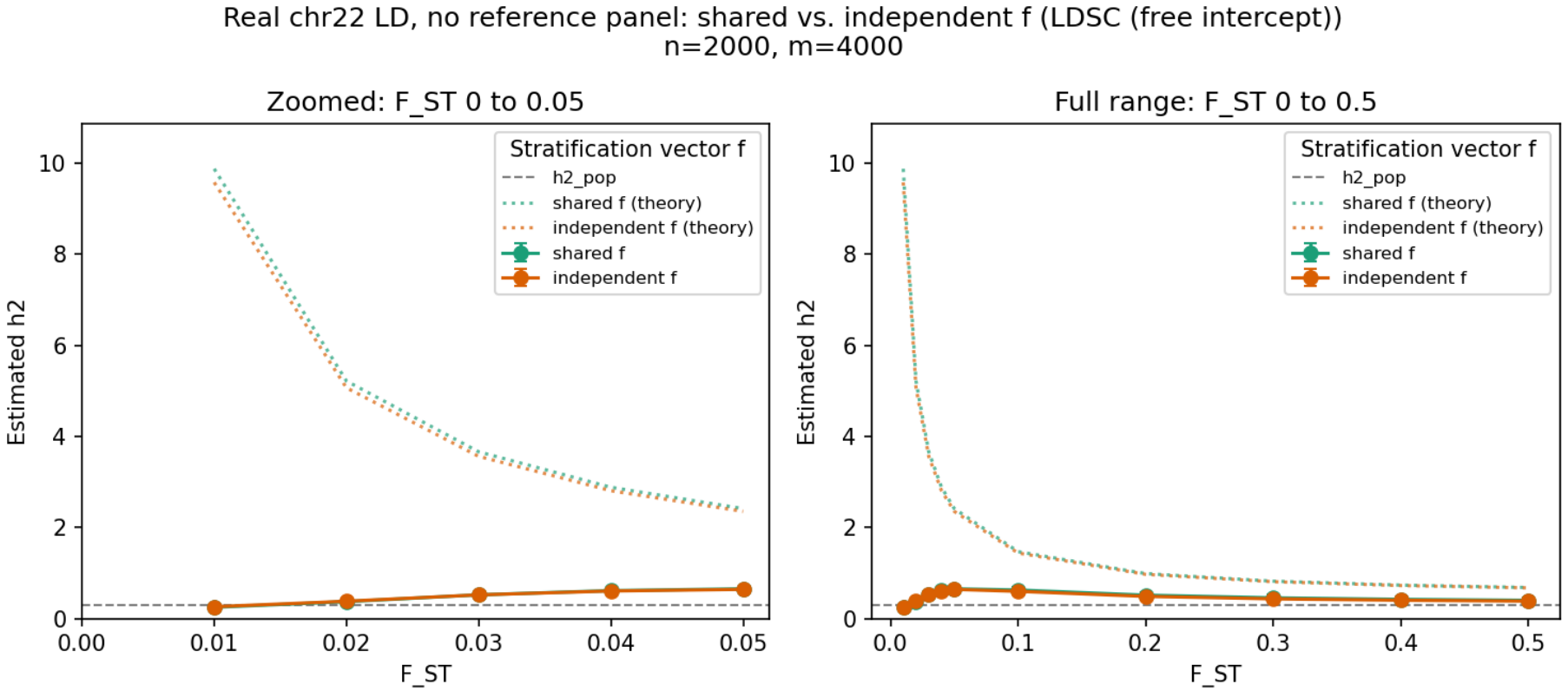
chr1, n=1000, m=2000 — no reference panel



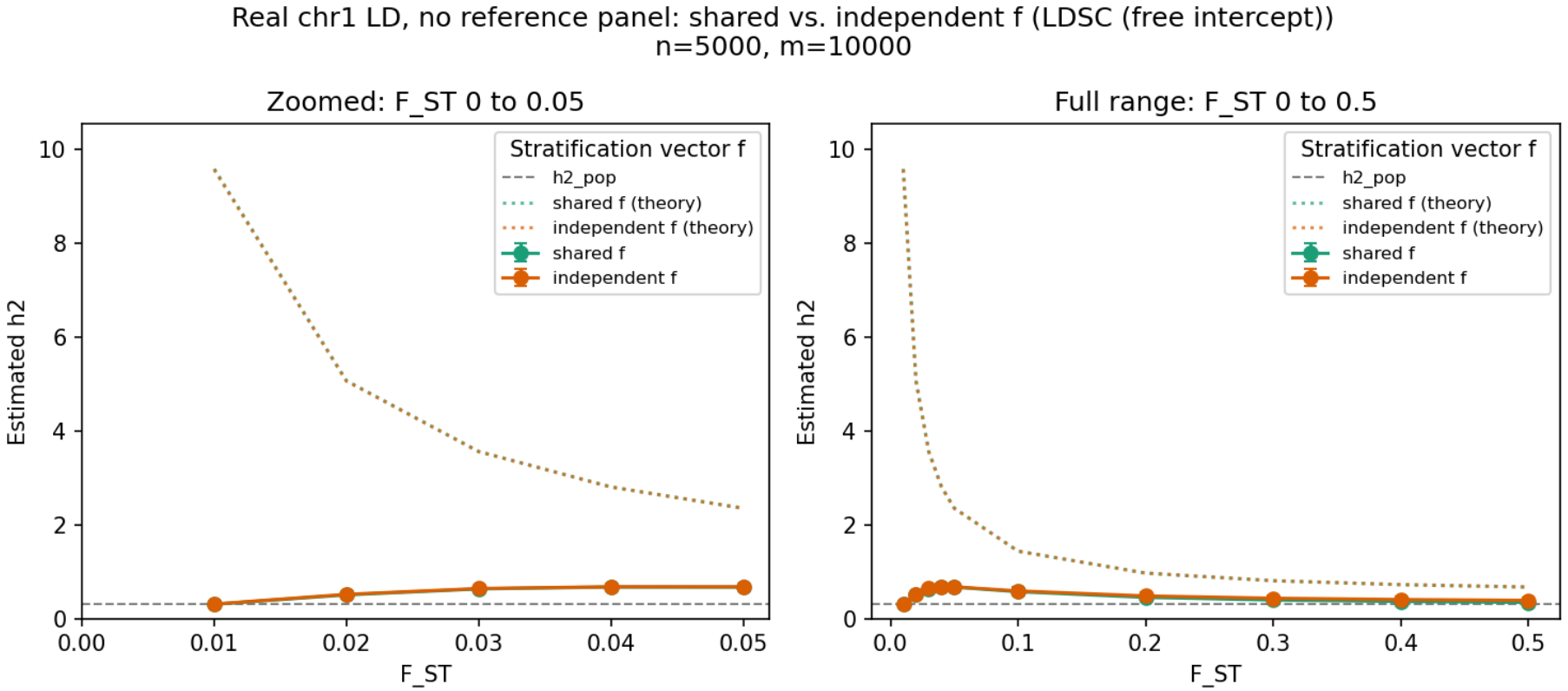
chr22, n=1000, m=2000 — no reference panel



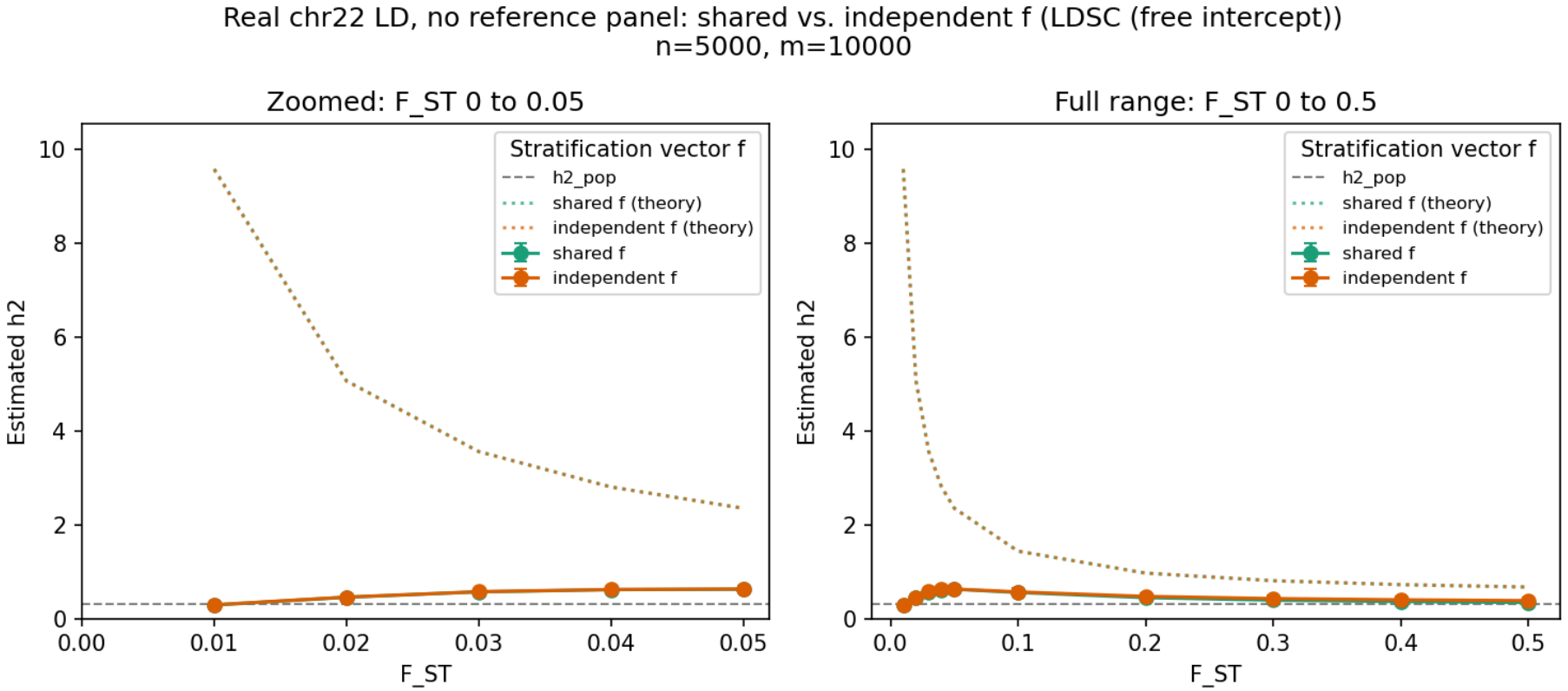
chr1, n=2000, m=4000 — no reference panel



chr22, n=2000, m=4000 — no reference panel



chr1, n=5000, m=10000 — no reference panel



chr22, n=5000, m=10000 — no reference panel

Discussion

Why no-ref-panel is robust to f-sharing: when LD scores and GWASH moments are recomputed in-sample every replicate, the estimator always sees a self-consistent sample - whether that sample's own f happens to be fixed across replicates or redrawn each time is irrelevant to any single replicate's estimate. This is confirmed across all four scales and both chromosomes.

Why ref-panel independent-f gets worse with scale: a mismatched reference panel supplies LD scores computed from a *different* realization of population structure than the main sample's. As n and m grow, the main sample's own signal becomes more precisely estimated (tighter confidence intervals, visible in the figures), but it is being regressed against LD scores that reflect the wrong stratification pattern. The systematic bias this introduces does not shrink with more data - sample size reduces variance, not the bias from using a structurally mismatched reference panel - so the gap between shared-f and independent-f widens as the estimates themselves become more precise.

Inflation vs. collapse: this contrasts with the earlier AR1-based reference-panel finding, where an f-mismatched reference panel caused the free-intercept estimator to collapse toward zero rather than inflate. The sign of the failure mode appears to depend on the LD structure used (synthetic AR1 vs. real chromosome LD), which is itself a notable result: it is not safe to assume the direction of bias from a mismatched reference panel without knowing the underlying LD structure.

chr1 vs. chr22: the near-identical gap trajectories (within ~0.1-0.15 of each other at every scale) across two chromosomes of very different size (106,136 vs. 18,613 SNPs) and gene density suggest this is primarily a sample-size effect of the simulation design, not an artifact of a particular chromosome's LD architecture.

Practical implication: these results reinforce that a reference panel is only useful for this kind of heritability estimation if its stratification structure genuinely matches the target sample's - and that mismatches do not wash out with larger studies. If anything, larger reference-panel-based studies have more to lose from an f-mismatch than smaller ones.

Appendix: Data and Reproducibility

All 352 simulation result CSVs (11 F_{ST} values × 4 scenarios × 4 scales × 2 chromosomes, 100 replicates each) are saved under data/python_realistic_* (chr1) and data/python_realistic_*_chr22 (chr22) in the project repository, and are not regenerated by the plotting scripts - replotting or re-analysis can be done directly from these files.

Driver scripts (python/):

- do_realistic_sim_Sam_noref_sharedf[_chr22].py, do_realistic_sim_Sam_noref_indepf[_chr22].py
- do_realistic_sim_Sam_ref_fixedf[_chr22].py, do_realistic_sim_Sam_ref_indepf[_chr22].py
- make_ref_panel_realistic[_chr22].py
- plot_realistic_sharedf_vs_indepf[_chr22].py <n> (regenerates figures from saved CSVs only)

Each driver script accepts F_{ST} and an optional sample size n (m = 2n, default n=500) as command-line arguments.